**Tableau Chap 2: Analyzing Data**

**Lesson 1: Preparing For Analysis**

***Analyzing Data in Tableau* (DataCamp — Lis Sulmont)**

### 1. Data Preparation — Key Questions to Ask

Before diving into analysis, evaluate your dataset with these questions:

- **Do any fields need to be refined?** (naming, formatting, data types)
- **Are there calculated fields we can create** to more effectively tell our data story?
- **Does the data contain fields that allow for summaries or grouping** at a higher level?
- **Are there sufficient categorical fields** to **slice and dice** your data?

### 2. Case Study Dataset: Divvy Bike Share

**stations table**

| Field | Description |
|---|---|
| id | ID attached to each station |
| name | Station name |
| latitude | Station latitude |
| longitude | Station longitude |
| docks | Number of docks at the station |

**trips table**

*(Trips taken between Jan–June 2019)*

| Field | Description |
|---|---|
| trip_id | ID attached to each trip |
| bikeid | ID attached to each bike |
| tripduration | Trip time in seconds |
| starttime | Day/time trip started (CST) |

| | |
|---|---|
| endtime | Day/time trip ended (CST) |
| from_station_id | Station ID of trip start |
| from_station_name | Station name of start |
| to_station_id | Station ID of trip end |
| to_station_name | Station name of end |
| usertype | customer or subscriber |
| birthyear | Birth year of rider |
| gender | Gender of rider |

**3. Dimensions & Measures Recap**

| | **Definition** |
|---|---|
| **Dimension** | Categorical or qualitative data |
| **Measures** | Numerical data that can be aggregated |

**Key strategy:** Move fields strategically between the two types.

- e.g., move numeric fields that **shouldn't be aggregated** (like an ID number, year, or zip code) into the **Dimensions** section — since summing/averaging them wouldn't make analytical sense.

**4. Calculated Fields**

Used to **extend the data** — creating new fields derived from existing ones to support deeper or more customized analysis (not just using raw columns as-is).

**5. Visualizations for Trend Analysis**

**Why look for trends?**

Trends inform business decisions such as:

- Marketing opportunities
- Scheduling maintenance
- Managing size/scheduling of staff

- Increasing or decreasing product stock/availability
- Trends can appear at different granularities: **hourly, daily, monthly, annual**

**Discrete vs. Continuous Time Analysis**

| Type | Description | Example Use |
|---|---|---|
| **Discrete (bins)** | Data grouped into categorical time buckets | Trends by hour, day of week, month |
| **Continuous (time series)** | Data shown in the exact sequence it historically occurred | Full trendline over time (e.g., 2016–2019) |

**Configuring dates in Tableau:** Right-click a date field (e.g., HOUR(Start Time)) to change the level of granularity (Year, Quarter, Month, Day, Hour, Minute, Week Number, etc.) or switch to **"Exact Date."**

**Chart Types Compared**

- **Trends Over Time (Bar Chart)** — shows discrete values per period (e.g., month), useful for comparing individual periods.
- **Trends Over Time (Line Chart)** — same data, but shows the continuous flow/direction across time — better for spotting overall trajectory.
- **Monthly Pattern (Bar Chart, sorted)** — aggregates all years together by month and sorts descending — reveals **seasonality** (e.g., Nov/Dec/Sep as peak sales months).
- **Monthly Trendline** — aggregated monthly average/sum shown as a single continuous line — highlights the seasonal pattern more clearly than bars.

**Continuous vs. Discrete side-by-side:** Continuous shows every individual month across every year in sequence (long-term pattern); discrete/aggregated view collapses all years into 12 monthly buckets (seasonality pattern).

### 6. Discrete Time Analysis & Quick Table Calculations

*(Section introduced — builds on the discrete/continuous distinction to apply Tableau's built-in Quick Table Calculations, e.g., running totals, percent of total, moving averages, for deeper trend analysis.)*

### 7. Slicing and Dicing

*(Final section — using categorical dimensions to break down measures into meaningful subgroups, tying back to the opening question: "Are there sufficient categorical fields to slice and dice your data?")*

### 🔑 Quick Takeaways

1. Always audit your data first — refine fields, spot calculation opportunities, check for grouping/categorical richness.
2. Correctly classify **dimensions vs. measures** — don't let non-aggregatable numbers sit in Measures.
3. Calculated fields let you extend raw data into more useful analytical fields.
4. Choose **bar charts** for discrete comparisons, **line charts** for continuous trends.
5. Aggregating across years by month reveals **seasonality** that a straight timeline might hide.
6. Adjust date granularity directly in Tableau's field menu (Year → Quarter → Month → Day → Hour, etc.).

**Lesson 2: Exploring Visualizations**
**Lesson 3: Mapping Analysis**
**Lesson 4: Group, Sets, and Parameters**